Supplementary Table S5. Architecture and temporal modelling of the 69 included studies

Study	Architecture	Model name	Temporal component	Task
S01 — Wang et al. (2023)	GCN-LSTM with attention-based meteorological fusion	NR	LSTM; attention	Multi-site PM2.5 forecasting
S02 — Zhou et al. (2024)	Multi-scale Spatiotemporal Graph Attention Network	MSTGAN	Transformer; GRU; convolution	Air-quality forecasting; PM2.5 forecasting; AQI prediction
S03 — Huo et al. (2026)	Continuous Spatio-Temporal Hybrid Graph Convolutional Network	CSHGCN	GRU; Neural ODE	Multi-city air-pollutant prediction
S04 — Jana et al. (2024)	Spatio-Temporal Graph Convolutional Network variants STGCN-A, STGCN-B, and STGCN-C	STGCN-C (principal/final architecture in the paper's overall conclusion)	Gated temporal convolution / 1D temporal convolution	Short-term multi-pollutant air-quality forecasting
S05 — Qi et al. (2019)	Graph Convolutional Neural Network + Long Short-Term Memory	GC-LSTM	LSTM	Multi-step PM2.5 forecasting
S06 — Xiao et al. (2022)	Dual-Path Dynamic Directed Graph Convolutional Network	DP-DDGCN	GRU	Multi-step PM2.5 forecasting
S07 — Sui and Han (2023)	Multi-view Multi-task Spatiotemporal Graph Convolutional Network	M^2	GRU encoder; LSTM decoders; FFT/IFFT denoising	Multi-step PM2.5 forecasting with auxiliary PM2.5 pollution-level prediction
S08 — Teng et al. (2023)	Hybrid graph deep neural network + LSTM	GNN_LSTM	LSTM	Real-time multi-step PM2.5 forecasting; source-receptor/transport analysis
S09 — Zhou et al. (2021)	Dynamic Directed Spatio-Temporal Graph Convolutional Network	DD-STGCN	Temporal convolution	Multi-step PM2.5 forecasting
S10 — Zhou et al. (2022)	Theory-guided graph network (extended DPGN framework)	NR	LSTM	Multi-step PM2.5 forecasting; source-receptor/causal analysis
S11 — Chen et al. (2025)	Hybrid directed GNN-LSTM framework using GAT-LSTM modules	GL-GL	LSTM	Regional multi-step PM2.5 forecasting using historical and forecast meteorology
S12 — Guyu et al. (2025)	Mixed Graph Convolutional GRU + Self-Attention Network	MGCGRU-SAN	GRU; self-attention	Multi-step PM2.5 forecasting
S13 — Shi et al. (2026)	Adaptive Spatiotemporal Graph Transformer	AST-GT	Transformer; temporal self-attention; TSCA gating; nonstationary normalization	Multi-site multi-step PM2.5 forecasting; sparse-station robustness/spatial extrapolation
S14 — Zhao et al. (2026)	GCNGRU with Fuzzy Tsallis Entropy station-importance branch	FTE-GCNGRU	GRU	Multi-step PM2.5 forecasting; key-station importance analysis
S15 — Ni et al. (2023)	Adaptive Scalable Spatio-Temporal Graph Convolutional Network	ASGCN	Customized temporal convolution network with gated multi-kernel convolutions	Single-step and multi-step PM2.5 forecasting
S16 — Kim et al. (2023)	Multi-gated spatiotemporal graph neural network with graph attention	NR	GRU; temporal encoding	Mid-to-long-term multi-step PM2.5 forecasting
S17 — Zeng et al. (2025)	Dynamic Adaptive Graph-generating Jumping Network	DAGJN	Multi-head self-attention	Multi-step air-quality forecasting; primary PM2.5 prediction with additional pollutant experiments
S18 — Liu et al. (2026)	Graph-integrated latent ensemble Random Vector Functional Link without Direct Links	AGILE-RVFLwoDL	Online ensemble/recursive least squares with forgetting; adaptive source weighting; context-aware gating	Drift-resilient online direct multi-step PM2.5 forecasting; cross-city online transfer
S19 — Xu et al. (2023)	Dynamic Graph Neural Network with Adaptive Edge Attributes	DGN-AEA	GRU	Multi-step PM2.5 forecasting
S20 — Ma et al. (2026)	Dynamic multi-graph ensemble GNN framework; graph-specific predictors use gated TCN, hybrid graph processing and adaptive spatiotemporal attention; RLMC uses a DDPG actor–critic for graph-level integration	DMGENet	Gated temporal convolutional network (TCN); adaptive spatiotemporal attention	Multi-station multi-step PM2.5 forecasting across four regional datasets
S21 — Su et al. (2023)	Multiple spatial–temporal GNN with two GCN graph views and temporal-attention GRU	GCNTAG	GRU; temporal attention	48-h PM2.5 forecasting and PM2.5 prediction in areas without monitoring stations
S22 — Zhao et al. (2025)	Spatial-Temporal Hypergraph Attention Network; graph component combines Hypergraph Attention (HGAT) and GAT	STHGAT	GRU	PM2.5 forecasting
S23 — Gao et al. (2026)	Adaptive Hypergraph-based Convolution Network with Dual Spatiotemporal Attention; graph component combines adaptive GCN and adaptive HGCN with feature fusion	AHCN-DA	TCN with dilated causal convolutions; spatial and temporal attention	One-step PM2.5 forecasting
S24 — Zhang et al. (2024)	ECGAT-GCFC hybrid; graph component = GNN + Edge Channel + GATConv	ECGAT-GCFC (EGCFC)	Gating Closed-form Continuous-time Neural Network (GCFC); GLU gating	Long-term multi-step PM2.5 forecasting
S25 — Mandal and Thakur (2023)	Spatially Attentive Cluster-based GNN; graph component uses edge-MLP/node-MLP plus GAT within station clusters	SA-GNN	Cluster-wise GRU; 1D-CNN + GRU temporal module	Multi-step PM2.5 forecasting; spatial PM2.5 estimation maps
S26 — Hu et al. (2024)	Adaptive Hierarchical Graph Convolutional Neural Network; graph component = hierarchical multi-graph convolution with attention-based multi-level aggregation	AHGCNN	AHGC-GRU encoder/decoder; GRU	Multi-step PM2.5 forecasting
S27 — Tan et al. (2022)	Ensemble graph-attention spatiotemporal predictor combining GAT-LSTM and GAT-TCN with reinforcement-learning ensemble control	GAT-LSTM-TCN-Sarsa	LSTM; TCN; Sarsa reinforcement-learning ensemble	Spatiotemporal PM2.5 prediction
S28 — Gao and Li (2021)	Graph-based Long Short-Term Memory with a learned parameterized adjacency embedded in LSTM information aggregation	GLSTM	LSTM	Synchronous all-station one-step and multi-step PM2.5 forecasting
S29 — Liu et al. (2021)	Graph convolutional network for spatial feature extraction, followed by GCN-LSTM and GCN-GRU predictors and reinforcement-learning ensemble selection	GCN-LSTM-GRU-Q	LSTM; GRU; Q-learning ensemble	Spatiotemporal PM2.5 forecasting
S30 — Zhang et al. (2024)	PMPre-GCN: graph convolution for spatial correlation plus spatiotemporal convolution blocks with gated temporal convolution	PMPre-GCN	Gated temporal convolution; GLU; temporal convolution	Fine-grained multi-station PM2.5 forecasting
S31 — Ma et al. (2022)	Spatio-Temporal Graph Convolutional Network with spatial GCN and gated temporal graph-convolutional blocks	STGCN-K4 / STGCN-ALL (principal all-observation setting)	Gated temporal convolution / graph convolutional temporal module	Spatiotemporal air-quality/PM2.5 prediction using fused heterogeneous monitoring data
S32 — Gu et al. (2025)	Dynamic wind-driven graph fusion model with graph attention and encoder–decoder recurrent temporal modelling	NR — authors refer to it as the dynamic wind-driven graph fusion model	Bidirectional GRU encoder; GRU decoder; self-attention/additive attention	Multi-step PM2.5 forecasting; transfer learning across stations; multi-pollutant transfer experiment
S33 — Zhang et al. (2024)	Dynamic Multi-Graph Attention module followed by LSTM	MGAtt-LSTM	LSTM	Multi-step PM2.5 forecasting
S34 — Zhao et al. (2025)	Graph Convolution Long Short-Term Memory using Dynamic Geographical Information; dynamic graph convolution + LSTM	DGGNN	LSTM	Multi-step PM2.5 forecasting
S35 — Zhang et al. (2025)	Multi-granularity temporal decomposition/SCIBlock combined with spatiotemporal graph construction and Graph Attention Network	NR — authors describe it as a multi-granularity PM2.5 concentration long-sequence prediction model	SCIBlock/SCINet-style multi-resolution temporal modelling; 1D convolution; interactive learning	Long-sequence multi-city PM2.5 forecasting
S36 — Ye et al. (2025)	GNN-based Spatial Impact Model plus Transformer-based Spatiotemporal Prediction Module, preceded by an IPS estimator	GNN-Transformer	Transformer; ProbSparse/self-attention-style spatiotemporal prediction	Multi-step PM2.5 forecasting
S37 — Velez-Serrano and Alvaro-Meca (2026)	Improved Spatio-Temporal Graph Transformer: residual GCN layers + dilated TCN blocks + Transformer encoder + attention pooling; final stacking combines ISTGT and MTSP forecasts using CatBoost	ISTGT; final stacking model = ISTGT + MTSP	Dilated TCN; Transformer encoder; temporal attention pooling; MTSP/SARIMA component in final stack	One-day-ahead municipal PM2.5 forecasting; spatial extrapolation to unsampled grid locations
S38 — Dong et al. (2024)	Uncertainty-guided graph convolution recurrent neural network: uncertainty-guided graph convolution module + spatial information modulator embedded in GRU encoder–decoder	AirNN	GRU encoder–decoder	Multi-step air-quality forecasting across monitoring-station networks
S39 — Dun et al. (2022)	Dynamic graph convolution neural network with multi-channel temporal convolutional network	DGC-MTCN	Multi-channel TCN; causal/dilated temporal convolution; residual blocks	Single-step and multi-step PM2.5 forecasting

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S5 continued

Study	Architecture	Model name	Temporal component	Task
S40 — Tariq et al. (2023)	Distance Adaptive Graph Convolutional Gated Network	DAGCGN	Gated recurrent network / GRU-style gated cells	Multi-site multi-pollutant forecasting; long-term sensor-failure imputation; remote sensing at sensor-devoid locations; CAI health-risk prediction
S41 — Wang et al. (2024)	Graph Attention Network + Gated Recurrent Unit	GAT-GRU	GRU	PM2.5 and O3 forecasting; composite pollution analysis; regional collaborative-control interpretation
S42 — Zhang et al. (2024)	Spatiotemporal Graph Attention Recurrent Network optimized by Grey Wolf Optimization; multi-GAT + multi-GNN + GRU	GWO-GART	GRU	Long-term multi-step PM2.5 forecasting
S43 — Liu et al. (2024)	Comparative graph-learning benchmark using GCN; STGCN; ASTGCN	NR	GCN: none; STGCN: temporal/1D convolution; ASTGCN: temporal attention	24-h-ahead PM2.5 forecasting under random versus temporal distribution shift
S44 — Mohammadzadeh et al. (2024)	Integrated Enhanced LSTM + Graph Convolutional Network	E-LSTM-GCN	Enhanced LSTM (E-LSTM)	Next-day PM2.5 AQI forecasting across six Michigan locations
S45 — Liao et al. (2024)	Spatiotemporal graph neural network based on the PM2.5-GNN framework with recurrent temporal modelling; a two-GNN simulation framework separates ambient and fire-related contributions	GNN-based prescribed-fire PM2.5 simulation framework	RNN	Hourly PM2.5 forecasting; fire-specific PM2.5 estimation; prescribed-fire scenario simulation
S46 — Han et al. (2025)	Comparative GCN/decomposition framework evaluating GCN; EMD+GCN; EEMD+GRU+GCN; and GCN with attention	Unclear from full text — contributions emphasize EEMD+GRU+GCN_ATT, while result tables/conclusion refer to EEMD+GRU+GCN	EMD/EEMD decomposition; GRU in the EEMD+GRU+GCN variant	Multi-pollutant air-quality forecasting for PM2.5, NO2 and SO2
S47 — Huang et al. (2025)	Graph Convolutional Network + FFPformer (Transformer variant with frequency-domain processing)	GCN-FFPformer	Transformer encoder/decoder; Fast Fourier Transform; probabilistic attention	Long-term multi-step PM2.5 forecasting
S48 — Wang et al. (2024)	Graph LSTM encoder–decoder integrating GCN and LSTM, with multi-head attention in the decoder	GLSTMMA	LSTM; multi-head attention	Multi-step multi-pollutant air-quality forecasting
S49 — Dimri et al. (2024)	Adaptive Graph Attention Network integrated with Temporal Convolutional Networks in an encoder–decoder forecasting/bias-correction framework	AGATNet	Temporal Convolutional Network; causal dilated 1D convolution	Bias correction of CMAQ 72-h PM2.5 forecasts
S50 — Qi et al. (2024)	Graph Attention Network for spatial features + recurrent multi-period temporal feature extraction	GMC (GAT-MULCYCLE)	GRU; multiple time-period/cycle feature extraction	Multi-source spatiotemporal PM2.5 prediction
S51 — Pan et al. (2024)	Attention-based graph neural network + gated recurrent unit	GARNN	GRU	Multi-city 72-h PM2.5 forecasting
S52 — Jin et al. (2023)	GraphSAGE spatial module + Bayesian Graph GRU temporal module with Bayesian hyperparameter optimization	BGGRU	BGraphGRU / GRU	One-step PM2.5 forecasting
S53 — Zhang et al. (2022)	Temporal Difference-Based Graph Transformer Networks with graph-attention encoder and decoder	TDGTN	Transformer; graph/self-attention	Single-step and multi-step PM2.5 forecasting
S54 — Zeng et al. (2022)	Spatial-Temporal Graph Ordinary Differential Equation Network with external humidity feature fusion	STGODE-M	TCN; neural ordinary differential equation / tensor ODE	Short-term PM2.5 forecasting
S55 — Lu et al. (2025)	Multi-Scale Dynamic Graph Neural Network with station grouping, multi-head/spatiotemporal graph attention, and Dynamic Adaptive Graph Convolution Network	MSDGNN	Multi-scale hourly/daily/weekly temporal modelling; multi-head self-attention	Multi-step PM2.5 forecasting
S56 — Abbasi et al. (2025)	Clustered lightweight Spatio-Temporal Graph Convolutional Network with GRU	ClusLite-STGCN-GRU	GRU	Multi-horizon multi-station PM2.5 forecasting
S57 — Guan et al. (2025)	Improved Graph Convolutional Network + GRU seq2seq + spatiotemporal attention; DRWF graph filter	STGCA	GRU encoder–decoder/seq2seq; attention	Regional high-resolution multi-step PM2.5 forecasting
S58 — Chen et al. (2024)	Adaptive graph convolution embedded in an LSTM-based spatiotemporal recurrent architecture	AAMGCRN	LSTM	Multi-step multi-pollutant air-quality forecasting at target monitoring stations
S59 — Chauhan et al. (2026)	Feature-attention module + Graph Attention Network + GRU	CIST-GNN	GRU	Next-step multi-station PM2.5 forecasting with feature-level interpretability
S60 — Li et al. (2023)	MLP feature extraction + Chebyshev GCN + Informer	GCNInformer	Informer/Transformer; ProbSparse self-attention	Multi-horizon multi-site air-quality forecasting
S61 — Mirzaei and Torghabeh (2026)	Neural Controlled Differential Equation + Graph Attention Network hybrid	Hybrid CDE + GAT / Neural CDE-GAT	Neural CDE; cubic-spline interpolation; RK4 integration	72-h multi-pollutant spatiotemporal forecasting; missing-data robustness
S62 — BADOUCH and BELHOUCINE (2026)	Two-layer GCNConv spatial encoder + Transformer encoder	GCN+Transformer (benchmark model; authors explicitly state the study does not propose a new architecture)	Transformer; self-attention	One-step PM2.5 forecasting on aggregated, single-station, and 12-node multi-station settings
S63 — Huang et al. (2025)	Multi-scale iTransformer + BSTA dynamic global graph + GESM local graph module	Dynamic global–local spatiotemporal graph framework	iTransformer/attention; GRU in local GESM module	Long-term multi-city PM2.5 forecasting
S64 — Pei et al. (2022)	Adaptive Variational Mode Decomposition + Multivariate Temporal GNN + GRU ensemble	PMNet	GRU; adaptive variational mode decomposition	Multi-step PM2.5 concentration forecasting
S65 — Yu et al. (2026)	Graph Attention Physics-Constrained Learning framework coupling multi-head GAT and Koopman-network latent dynamics	GAPCL	Koopman operator / physics-constrained latent linear dynamics	Short-term multi-station PM2.5 forecasting
S66 — Wang et al. (2024)	Variational Mode Decomposition + Graph Attention Network + Bidirectional LSTM	VMD–GAT–BiLSTM	BiLSTM; VMD decomposition	Short- and long-horizon PM2.5 forecasting; five-pollutant generalization analysis
S67 — Zhao (2025)	Spatial-Temporal Graph Convolutional Network	ST-GCN	Dilated causal temporal convolution	PM2.5 prediction; trend/spatial analysis; influencing-factor analysis; annual-average forecasting to 2030
S68 — Zeng et al. (2021)	Spatiotemporal attention + graph convolution + temporal convolution	STAM-STGCN	Temporal convolution; temporal attention	Short-term multi-station PM2.5 forecasting
S69 — Wang et al. (2022)	Graph Attention Network + Gated Recurrent Unit	GAT-GRU	GRU	72-h multi-city PM2.5 forecasting

Note: Architecture, proposed model name, temporal component, and task reproduce the study-level extraction. NR denotes not reported; category labels and tasks are not assumed to be mutually exclusive.